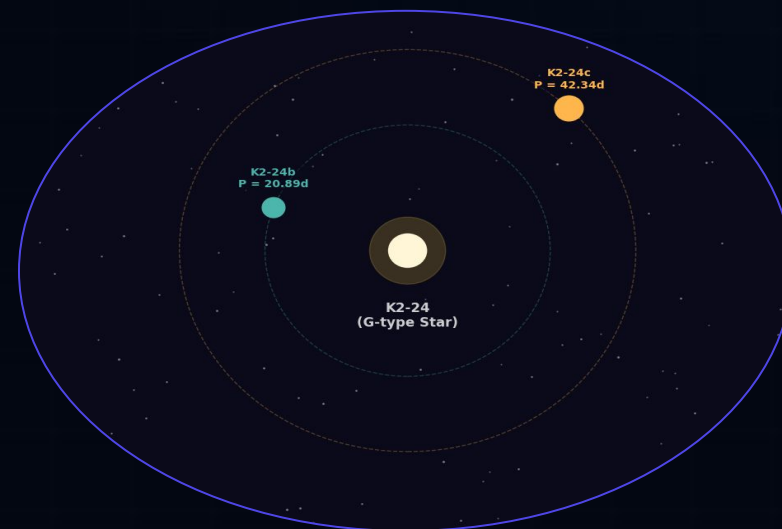


Deep Learning for Bayesian Computation

Applying Normalizing Flow Methods to Exoplanet Inference on K2-24

Daniel Christensen

Neural Inference | Exoplanet Orbital Dynamics



Why Exoplanet Inference?

Exoplanet astronomy has transitioned into an industrial-scale data challenge.

- ★ 5000+ confirmed exoplanets to date
- 📡 Radial velocity (RV) is a dominant indirect method
- 🔭 Posterior characterization defines habitability claims

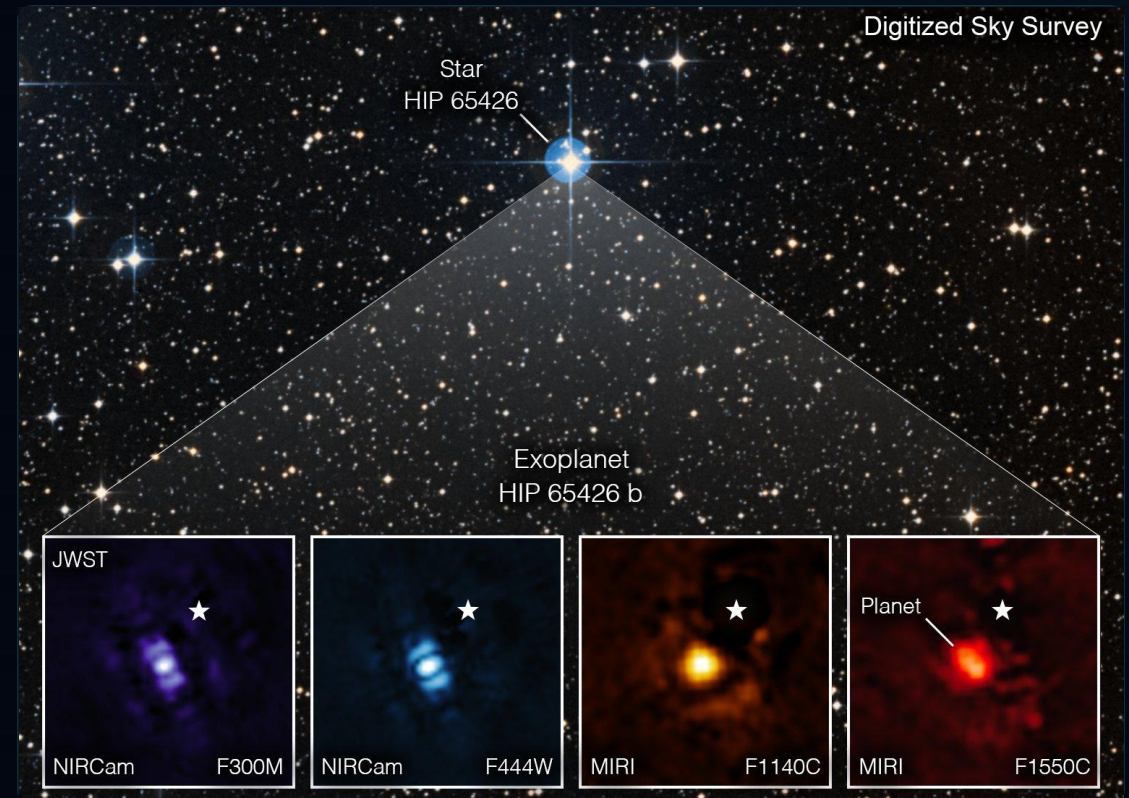


image of an exoplanet captured by the James Webb Space Telescope.

Credit goes to NASA/ESA/CSA, A Carter (UCSC), the ERS 1386 team, and A. Pagan (STScI).

PAIRED COMPARISON FRAMEWORK



HMC Baseline

Standard Hamiltonian Monte Carlo for exact but often slow sampling.



Flow-MCMC

Adaptive flow proposals to bridge multimodal gaps (Gabrié et al. 2022).



Mean-Field VI

Fast but restricted Gaussian approximation baseline.



Flow VI

Highly expressive variational family via Planar Flows (Rezende 2015).

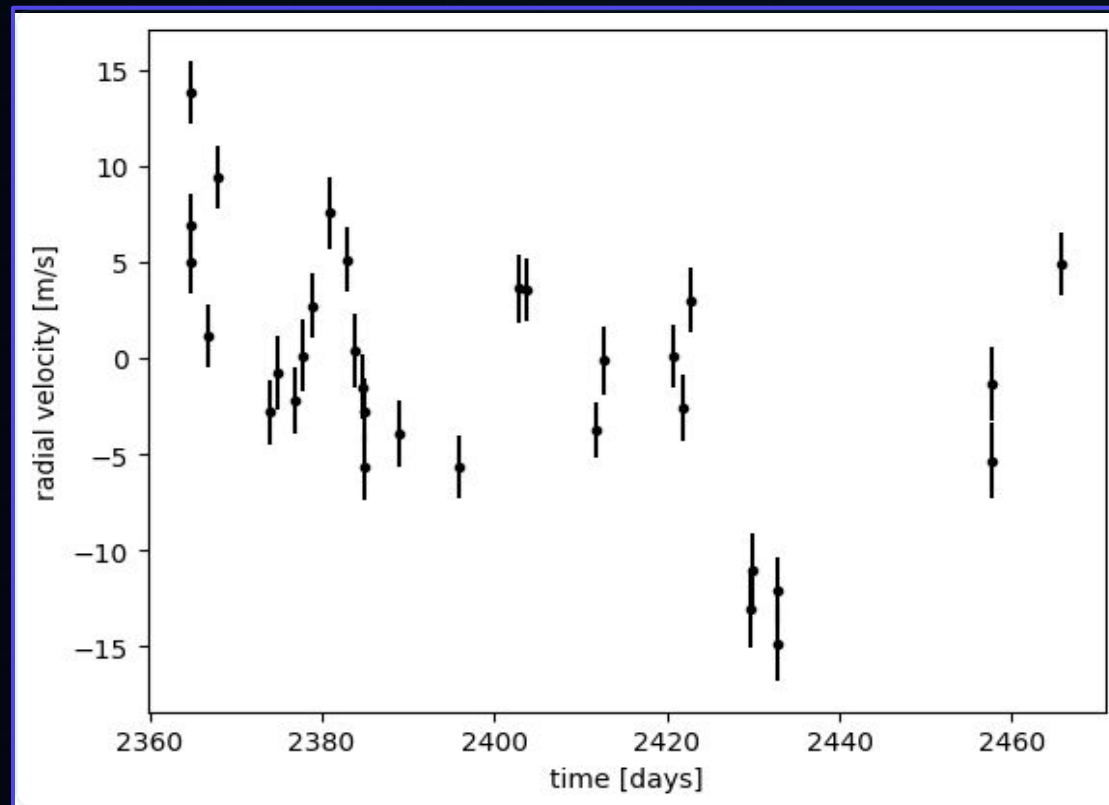
THE K2-24 DATASET

System Properties

EPIC 203771098: A Sun-like star with two known transiting Neptunian planets. Spans ~ 100 days, with two known transiting planets: $P_b = 20.89$ d, $P_c = 42.36$ d.

- 🗄️ 32 RV observations from Keck/HIRES
- 🕒 100-day observational baseline
- ✓ Ground truth periods from K2 photometry

32 RV observations from Keck/HIRES, EPIC 203771098



THE TWO-PLANET KEPLERIAN MODEL

$$v(t) = v_0 + \sum_{j=1}^2 K_j [\cos(\omega_j + \nu_j(t)) + e_j \cos \omega_j]$$

$$M_j(t) = \frac{2\pi}{P_j}(t - t_{p,j})$$

$$M_j = E_j - e_j \sin E_j$$

$$E^{(k+1)} = E^{(k)} - \frac{E^{(k)} - e \sin E^{(k)} - M}{1 - e \cos E^{(k)}}$$

$$\nu_j = 2 \arctan \left(\sqrt{\frac{1+e_j}{1-e_j}} \tan \frac{E_j}{2} \right)$$

$$\log p(\mathbf{v} \mid \boldsymbol{\theta}) = -\frac{1}{2} \sum_{i=1}^N \left[\frac{(v_i - v(t_i; \boldsymbol{\theta}))^2}{\sigma_i^2 + \sigma_J^2} + \log(2\pi(\sigma_i^2 + \sigma_J^2)) \right]$$

$$\boldsymbol{\theta} = (P_1, t_{p,1}, e_1, \omega_1, K_1, P_2, t_{p,2}, e_2, \omega_2, K_2, v_0)$$

🧠 **Differentiable:** Model implemented in PyTorch

🔻 **Jitter:** Absorbs unmodeled stellar activity

⚙️ **12 Parameters:** P, tp, e, omega, K (x2) + v0, jitter

🔗 **Newton Iteration:** To invert Kepler's equation

Weakly Informative Priors

-  **Log Period:** $\log P \sim \mathcal{N}(\log 50, 2^2)$ — Coverage: 7 to 370 days
-  **Periastron:** $t_p \sim \mathcal{N}(2400, 50^2)$ — Centered on data epoch
-  **Eccentricity:** $e \sim \text{Beta}(0.867, 3.03)$ — Calibrated via Kipping (2013)
-  **Semi-amplitude:** $K \sim \text{HalfNormal}(50)$ — Weakly informative for RV signal
-  **Log Jitter:** $\log \sigma_J \sim \mathcal{N}(0, 2^2)$ — Absorbs unmodeled stellar activity

$$\log P_j \sim \mathcal{N}(\log 50, 2^2)$$

$$t_{p,j} \sim \mathcal{N}(2400, 50^2)$$

$$e_j \sim \text{Beta}(0.867, 3.03)$$

$$K_j \sim \text{HalfNormal}(50)$$

$$v_0 \sim \mathcal{N}(0, 50^2), \quad \log \sigma_J \sim \mathcal{N}(0, 2^2)$$

Smooth priors avoid hard boundary gradients that could break VI optimization.

WHY THIS POSTERIOR IS HARD



High Dimensionality

12-D space precludes grid-based methods; requires intelligent exploration.



Multimodality

Period aliasing creates secondary modes at fractions of true periods.



Nonlinear Likelihood

Strongly nonlinear in P and e ; local curvature varies wildly across the space.

COORDINATE REPARAMETERIZATION

$$P_j = \exp(\eta_{P_j} \cdot 0.3 + \mu_{P_j})$$

$$t_{p,j} = \eta_{t_{p,j}} \cdot 50 + 2400$$

$$e_j = \sigma(\eta_{e_j} \cdot 1.5 + \mu_e)$$

$$\omega_j = \eta_{\omega_j}$$

$$K_j = \exp(\eta_{K_j} + \mu_{K_j})$$

$$v_0 = \eta_{v_0} \cdot 20, \quad \log \sigma_J = \eta_{\log J} \cdot 2$$

$$\log p(\boldsymbol{\eta} \mid \mathbf{v}) = \log p(\boldsymbol{\theta}(\boldsymbol{\eta}), \log \sigma_J(\boldsymbol{\eta}) \mid \mathbf{v}) + \log \left| \det \frac{\partial(\boldsymbol{\theta}, \log \sigma_J)}{\partial \boldsymbol{\eta}} \right|$$

$$\text{Period: } \log |d \log P / d\eta| = \log(0.3)$$

$$\text{Time of periastron: } \log |dt_p / d\eta| = \log(50)$$

$$\text{Eccentricity: } \log |de / d\eta| = \log(1.5) + \log e + \log(1 - e)$$

$$\text{Semi-amplitude: } \log |dK / d\eta| = \log K = \eta_K + \mu_K$$

$$v_0 : \log(20), \quad \log \sigma_J : \log(2)$$

Result: Posterior geometry becomes isotropic in "eta" space, allowing uniform MCMC step sizes.

HAMILTONIAN MONTE CARLO

$$H(\boldsymbol{\eta}, \mathbf{v}) = U(\boldsymbol{\eta}) + \frac{1}{2} \mathbf{v}^\top \mathbf{v}, \quad U(\boldsymbol{\eta}) = -\log \pi(\boldsymbol{\eta})$$

$$\frac{d\boldsymbol{\eta}}{dt} = \mathbf{v}, \quad \frac{d\mathbf{v}}{dt} = -\nabla U(\boldsymbol{\eta})$$

$$\mathbf{v}^{(t+\varepsilon/2)} = \mathbf{v}^{(t)} - \frac{\varepsilon}{2} \nabla U(\boldsymbol{\eta}^{(t)})$$

$$\boldsymbol{\eta}^{(t+\varepsilon)} = \boldsymbol{\eta}^{(t)} + \varepsilon \mathbf{v}^{(t+\varepsilon/2)}$$

$$\mathbf{v}^{(t+\varepsilon)} = \mathbf{v}^{(t+\varepsilon/2)} - \frac{\varepsilon}{2} \nabla U(\boldsymbol{\eta}^{(t+\varepsilon)})$$

$$\alpha = \min\left(1, \exp[H(\boldsymbol{\eta}^{(0)}, \mathbf{v}^{(0)}) - H(\boldsymbol{\eta}^{(L)}, \mathbf{v}^{(L)})]\right)$$

🌀 12 Chains | 500 samples each

🕒 Symplectic Leapfrog Integrator

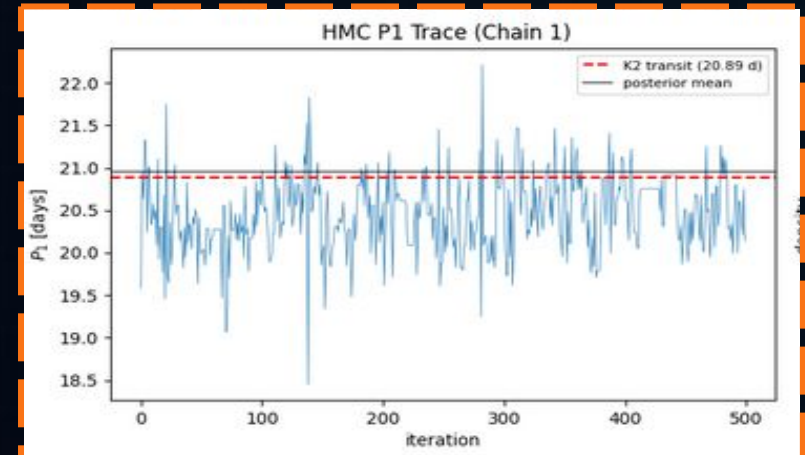
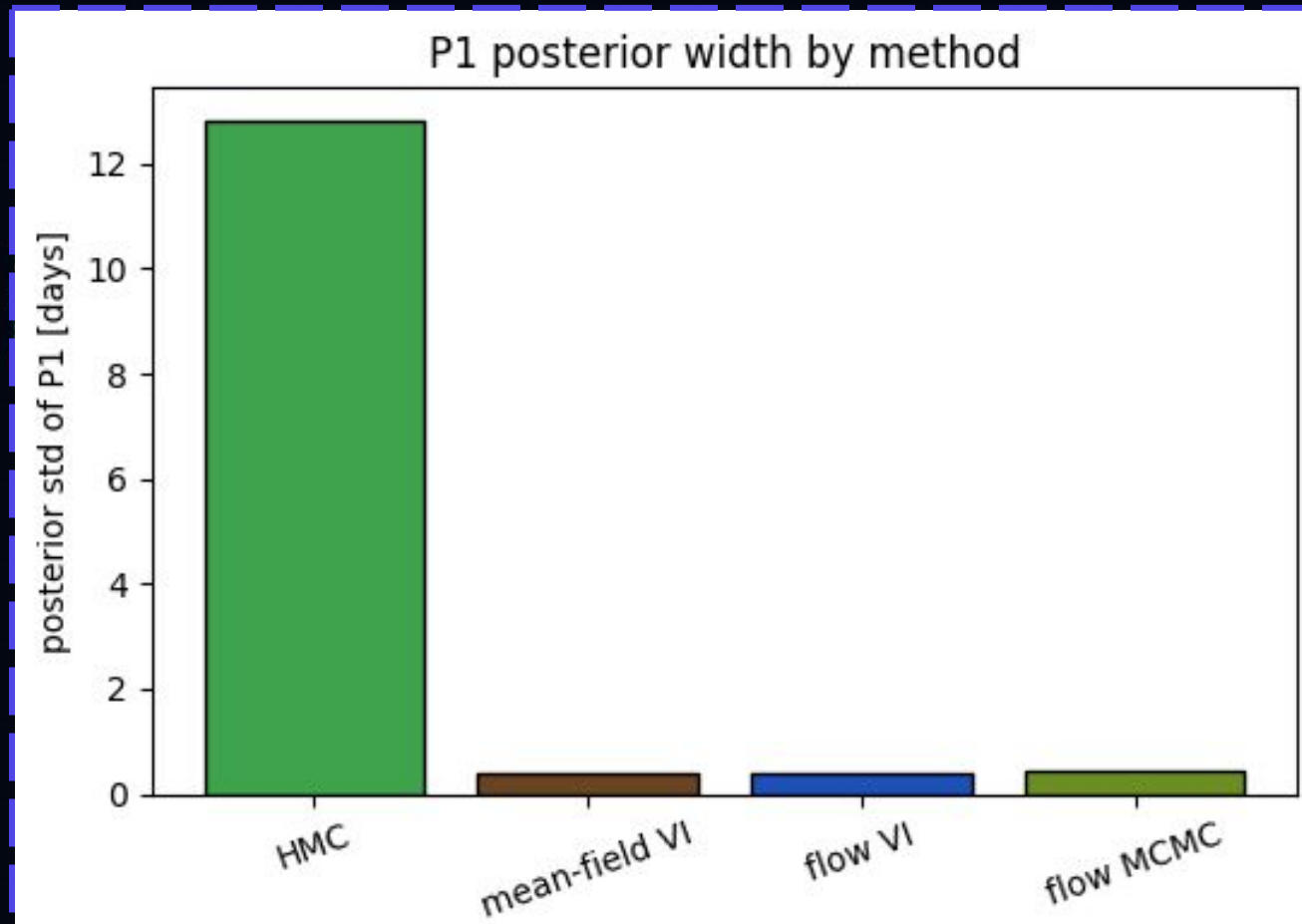
✓ Metropolis-Hastings correction

Augments posterior with auxiliary momentum variables.

Trajectories stay on energy level sets, allowing long steps while maintaining high acceptance rates.

Config: L = 30 steps, $\varepsilon = 0.01$.

HAMILTONIAN MONTE CARLO RESULTS



Chain 1: acceptance = 0.814
Chain 2: acceptance = 0.904
Chain 3: acceptance = 0.842
Chain 4: acceptance = 0.946
Chain 5: acceptance = 0.904
Chain 6: acceptance = 0.488
Chain 7: acceptance = 0.692
Chain 8: acceptance = 0.770
Chain 9: acceptance = 0.922
Chain 10: acceptance = 0.334
Chain 11: acceptance = 0.842
Chain 12: acceptance = 0.556

MEAN-FIELD VARIATIONAL INFERENCE

Optimizes the Evidence Lower Bound (ELBO) to approximate the posterior.

$$q^* = \arg \min_{q \in \mathcal{Q}} \text{KL}(q(\boldsymbol{\eta}) \parallel p(\boldsymbol{\eta} \mid \mathbf{v}))$$

$$\mathcal{L}(q) = \mathbb{E}_{q(\boldsymbol{\eta})} [\log p(\boldsymbol{\eta}, \mathbf{v}) - \log q(\boldsymbol{\eta})]$$

$$q(\boldsymbol{\eta}) = \prod_{i=1}^D \mathcal{N}(\eta_i; \mu_i, s_i^2)$$

$$\boldsymbol{\eta} = \boldsymbol{\mu} + \mathbf{s} \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(\mathbf{0}, I)$$

$$\hat{\mathcal{L}}(\boldsymbol{\mu}, \boldsymbol{\rho}) = \frac{1}{S} \sum_{s=1}^S \log p(\boldsymbol{\mu} + \mathbf{s} \odot \boldsymbol{\epsilon}^{(s)}, \mathbf{v}) + \mathbb{H}[q]$$

Mean-Field Factorization

$$q(\boldsymbol{\eta}) = \prod \mathcal{N}(\eta_i; \mu_i, s_i^2)$$

Assumes zero correlation between parameters. Uses the reparameterization trick for gradients.

Config: 2000 iterations, Adam @ lr=1e-2, 64 ELBO samples per gradient step, log_s init = -3

NORMALIZING FLOW-BASED VARIATIONAL INFERENCE

Planar Flows

$$\mathbf{z} \rightarrow \mathbf{z} + \mathbf{u} \cdot \tanh(\mathbf{w}^\top \mathbf{z} + b)$$

Captures complex correlations by warping a base Gaussian through 8 invertible planar layers.

Strictly better ELBO than Mean-Field.

Config: 8 planar layers, Adam @ lr=1e-3, 5000 iterations, 32 ELBO samples per step.

$$f(\mathbf{z}) = \mathbf{z} + \mathbf{u} \tanh(\mathbf{w}^\top \mathbf{z} + b)$$

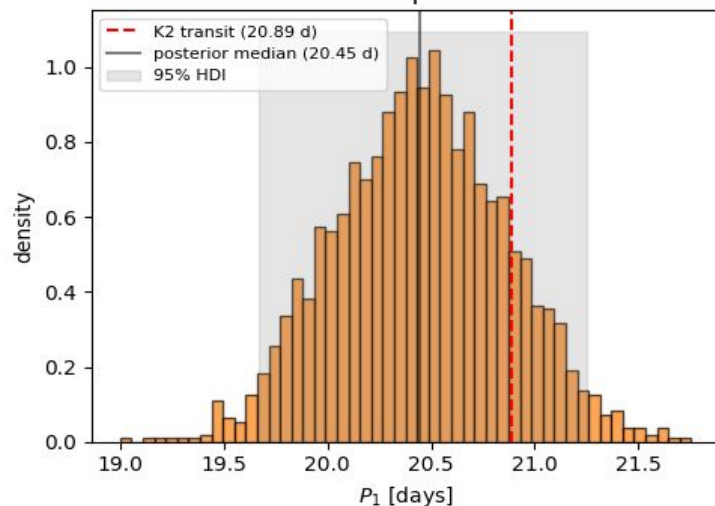
$$\log \left| \det \frac{\partial f}{\partial \mathbf{z}} \right| = \log |1 + \mathbf{u}^\top \psi(\mathbf{z})|, \quad \psi(\mathbf{z}) = (1 - \tanh^2(\mathbf{w}^\top \mathbf{z} + b)) \mathbf{w}$$

$$\hat{\mathbf{u}} = \mathbf{u} + [m(\mathbf{w}^\top \mathbf{u}) - \mathbf{w}^\top \mathbf{u}] \frac{\mathbf{w}}{\|\mathbf{w}\|^2 + \epsilon}, \quad m(x) = -1 + \log(1 + e^x)$$

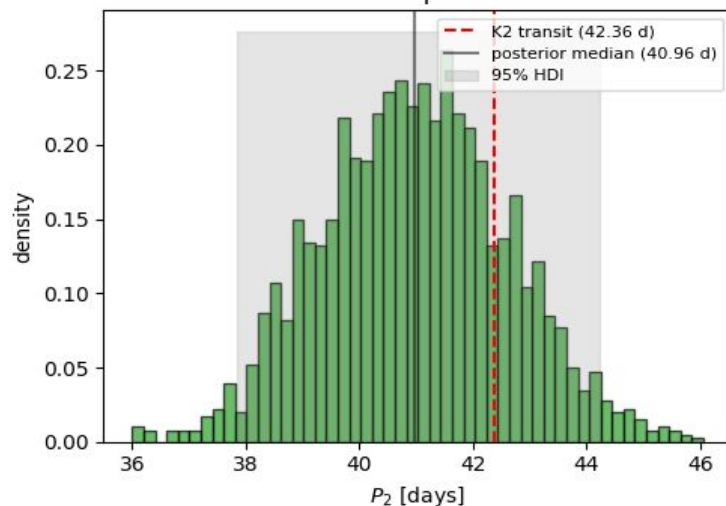
$$\log q_K(\mathbf{z}_K) = \log q_0(\mathbf{z}_0) - \sum_{k=1}^K \log \left| \det \frac{\partial f_k}{\partial \mathbf{z}_{k-1}} \right|$$

Variational Inference Methods Comparison

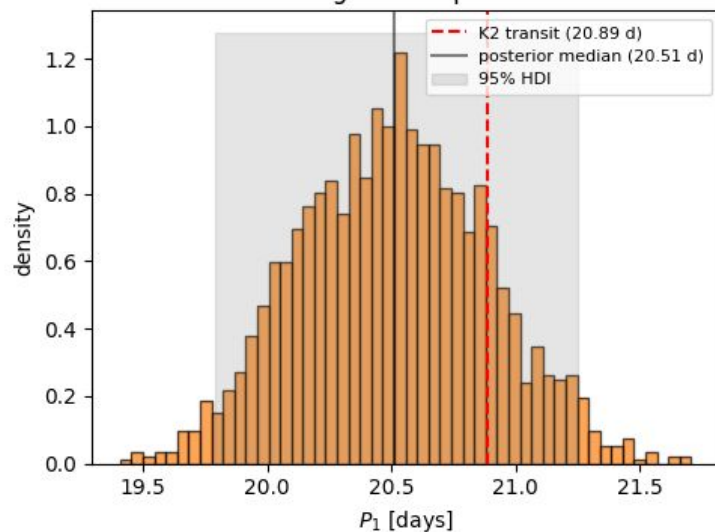
Mean-field VI posterior: P1



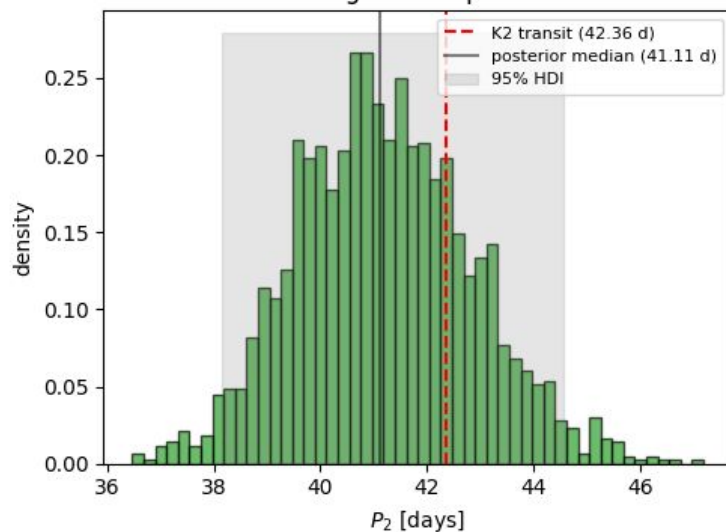
Mean-field VI posterior: P2



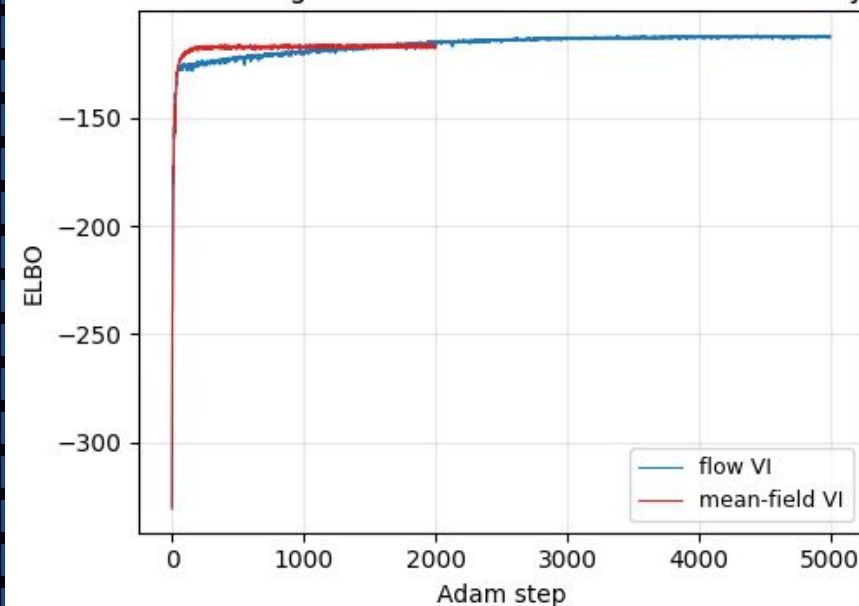
Normalizing Flow VI posterior: P1



Normalizing Flow VI posterior: P2



Normalizing Flow VI vs Mean-Field VI: ELBO history



Mean-field final ELBO = -116.49,

Flow VI final ELBO = -112.26,

$\Delta\text{ELBO} \approx 4$ nats,

P_1 std mean-field = 0.411

vs

Flow VI = 0.379.

Normalizing Flow MCMC

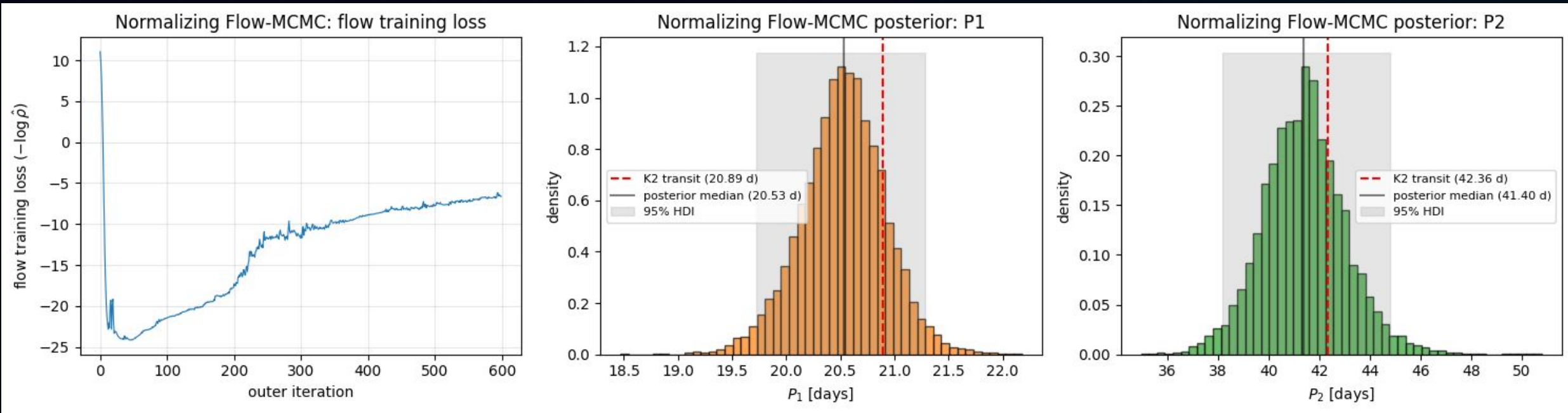
$$\eta_K = f_K \circ f_{K-1} \circ \dots \circ f_1(\eta_0)$$
$$\mathcal{L}_K = \mathbb{E}_{q_0(\eta_0)} \left[\log p(\eta_K, \mathbf{v}) - \log q_0(\eta_0) + \sum_{k=1}^K \log \left| \det \frac{\partial f_k}{\partial \eta_{k-1}} \right| \right]$$

Compound Kernel: Combines local MALA moves with global Flow proposals.

-  Real-NVP architecture (6 layers)
-  Adaptive training on chain history
-  Enables jumping between distant modes

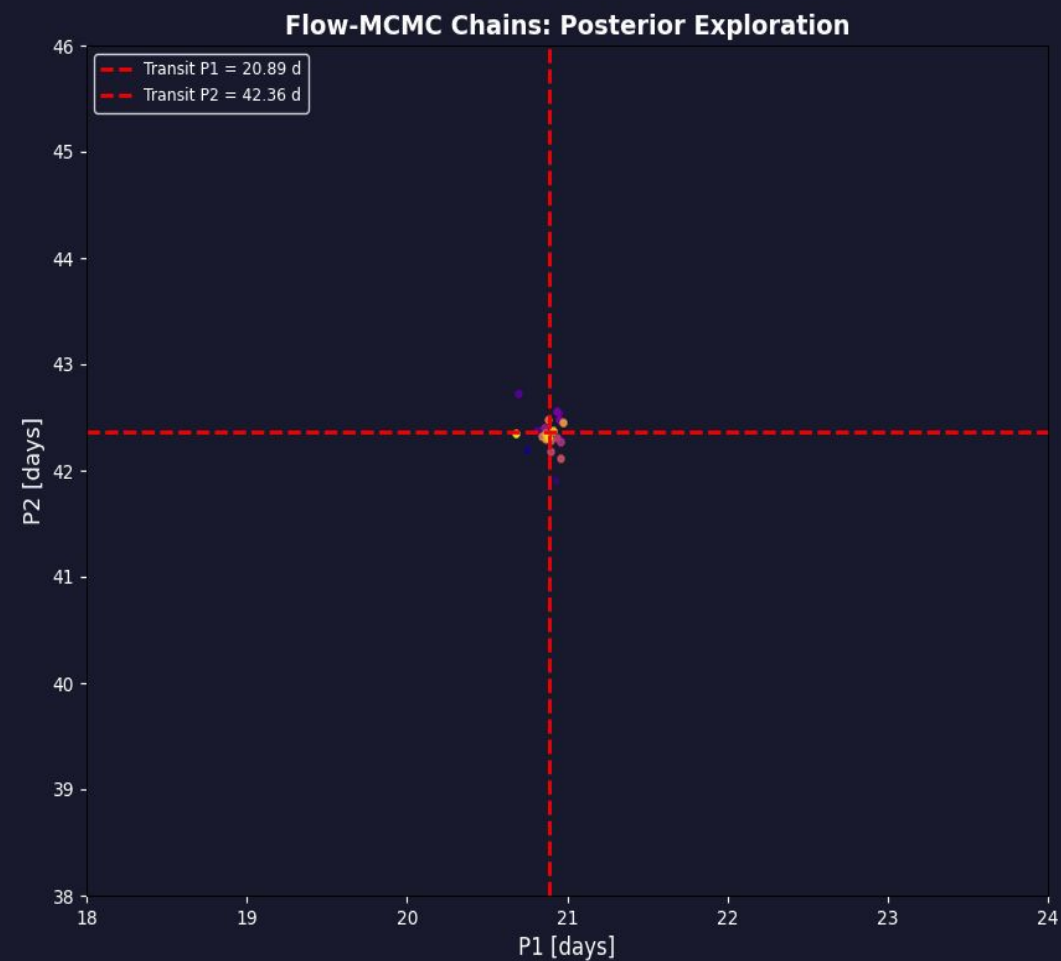
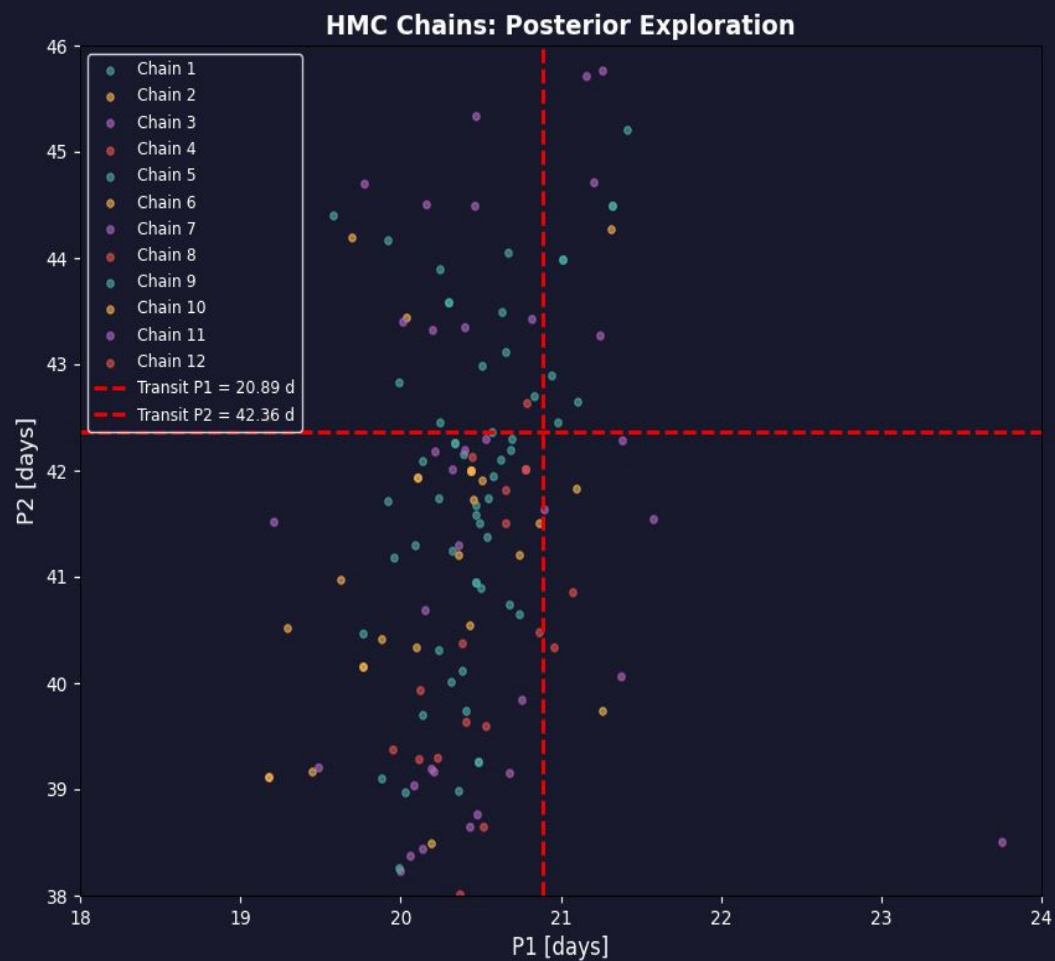
- Config: 24 chains × 600 outer iterations
- MALA $\tau = 5e-4$
- RealNVP with 6 layers and hidden=64, warmup of 50 iterations
- training buffer size = 2000.

Normalizing Flow MCMC Results



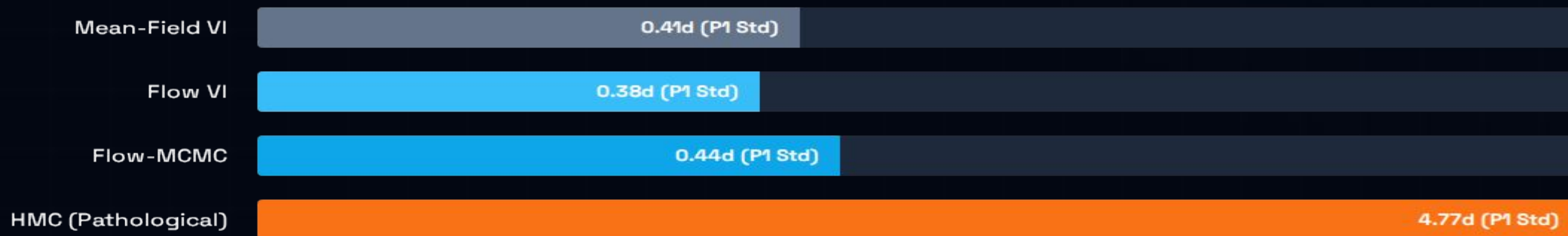
- Local MALA acceptance = 0.36
- Global flow acceptance = 0.02
- P2 std = 2.47 days
- All four sanity checks pass.

Side-by-Side Chain Comparison



CROSS-METHOD PERFORMANCE

Method	P1 std (day)	P2 std (day)	Final ELBO
Hamiltonian Monte Carlo	12.81	111.85	—
Mean-Field Variational Inference	0.41	1.63	-116.49
Normalizing Flow Variational Inference	0.38	1.65	-112.26
Normalizing Flow MCMC	0.44	2.47	—



VI offers confident but potentially over-confident unimodal fits. MCMC provides honest multimodal exploration.

SUMMARY & FUTURE DIRECTIONS

Core Findings

- ✓ Flow VI strictly improves on mean-field VI in ELBO (~ 4 nats), but mostly through joint correlations rather than marginal shapes
- ✓ Flow-MCMC produces more robust posteriors than vanilla HMC, with cleaner uncertainty estimates and chain-wandering protection
- ✓ Both flow methods are useful incremental improvements, not complete solutions

What's Next?

- ▶▶ Use Real-NVP as the Flow-VI backbone
- ▶▶ Tighter priors on semi-amplitude (K)
- ▶▶ R-hat Diagnostics across chains

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